Christopher Okoh

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Education

M.Sc. in Robotics Engineering
Liverpool, United Kingdom

Specialization: Robotics, Computer Vision, Reinforcement Learning and AI

Liverpool Hope Univeristy — *Sept 2025 - Oct 2026*

- **LHU Robotics Lab:** Graduate Research Assistant
- Masters thesis: **https://shorturl.at/r7HLS**
- Performed projects on sensor fusion, learning-based robot navigation, and vision-based robot control using **ROBOTNIK SUMMIT XL**

B.Eng. in Electrical and Electronics Engineering
Oye-Ekiti, Nigeria

Specialization: Intelligent control systems

Federal University Oye-Ekiti — *Feb 2018 - Jan 2023*

- Implemented adaptive controller design for a real-world 4-way traffic light using PLC and ultrasonic sensors
- Among the top 5% of best graduating students in the faculty

B.Tech in Physics
Owerri, Nigeria

Specialization: Electronic circuit/PCB design

Federal University of Technology, Owerri — *Dec 2003 - Oct 2010*

- Analog-Digital I/O: Integrated Circuit-based hardware projects
- Networking: System communication structure design and troubleshooting

Additional Training

Deep Learning Fundamentals Lab, Computer Vision Lab
Remote, Jul 2026

WorldQuant Univeristy

- Evaluated baselines for Multi-Layer Perceptron (MLP), CNN, and deeper architectures.
- Performed image classification, object detection/recognition, and gen AI for image synthesis.

Robot Student Showcase and Research Writing Training
Remote, Jul 2023

Frontiers in Robotics & AI

- Received training in academic article writing, peer-review process, and research dissemination.

Robot Operating System (ROS) Mentorship Programme
Remote, Aug 2022

RigBetel Labs

- Implemented SLAM and ROS autonomous navigation stack on custom URDF/SDF robot models.
- Explored the Gazebo simulation environment by performing navigation tasks with the **Tiago** robot

IVQ Level 5 Advanced Technician Diploma, Electrical and Electronics Engineering
IVQ Level 3 Technician Diploma in Electrical and Electronics Engineering
United Kingdom, Sept 2012 - Oct 2015

City & Guilds of London Institute, UK

- Assembled and programmed a four-wheel mobile robot using an Arduino microcontroller.
- Developed a PLC ladder logic program to control a level-flow test rig

Skills

Programming - Python (NumPy, PyTorch, TensorFlow, Keras, OpenCV, Matplotlib, Scikit-learn), C++, C#, ROS2.

Platforms - Linux (Ubuntu), Shell (Bash), Gazebo, Git, CoppeliaSim, RoboDK, Vizard 8, STM32, Arduino, ESP32, VS Code, Unity 3D.

Soft Skills - Teamwork, Time Management, Problem-solving, Documentation, Engaging Presentation.

Work Experience

Academic

Research Paper Reviewer - HRI 2026
Remote, Oct 2025 - Present

ACM/IEEE International Conference on Human-Robot Interaction (HRI)

- Provided careful assessments, detailed comments, and timely responses while maintaining high scholarly standards.
- Reviewed two research manuscripts, contributing to their successful acceptance, with one subsequently receiving the conference's Best Paper Award.

Teaching Assistant (TA)
Remote, Jul 2023 - Dec 2024

Black in Robotics (BiR)

- Delivered a range of teaching and assessment activities to twenty-five students on topics including Introduction to AI and Robotics with Raspberry Pi Pico.

- Led lab sessions on the control of a 3-DoF robotic arm using Python, achieving an average student pass rate of 80%

Summer School Intern - Robotics & AI
Barcelona, Spain, Jul 2022

Universitat Politecnica De Catalunya – Institut de Robotica I Informatica Industrial (Barcelona Tech)

- Developed mobile robot learning for traffic light color recognition and navigation control using SLAM and CNN.
- Explored an end-to-end learning baseline navigation system that includes collision avoidance during navigation to a target in a reduced traversal time.

Industrial

Robotics System Integration Engineer
PH, Nigeria, Oct 2024 - Sept 2025

Dufitek Institute of Applied Engineering and Technology

- Developed firmware for STM32 microcontrollers, utilizing C and STM32 IDE for low-level programming.
- Integrated communication protocols such as CAN, CAN-FD, and Ethernet for robot actuators.
- Designed and customized Docker environments to support scalable CI/CD pipelines.
- Collaborated with cross-functional teams to integrate motor control algorithms and robotic actuators

Drone Pilot
Onne, Nigeria, Sept 2023 - Oct 2024

Notore Chemical Industries

- Designed and simulated drone flight routes for adaptive aerial monitoring of critical infrastructures onshore/offshore.
- Planned and performed routine drone maintenance and firmware updates.

Research Assistant - Control Systems Engineer
PH, Nigeria, Mar 2014 - Jan 2018

CiTech Automation

- Integrated Simatic S7 and RSLogix with NI LabVIEW for data acquisition to develop a prototype control interface for a 3-DoF robot arm pick-and-place task.
- Designed and developed a ladder logic program for a flow and level control prototype rig system using Siemens S7 and Allen-Bradley MicroLogix1000 Programmable Logic Controllers (PLC).

Control Systems & Automation Technician
PH, Nigeria, Jan 2012 - Feb 2014

City Technique Engineering and Technical Services

- IInstalled F&G alarm systems, safety instrumented systems (SIL), and implemented safety instrumented functions (SIF) and safety integrity level (SIL).
- Executed the custom design of a proportional, integral, and derivative (PID) mechanism to control a level-flow process system.

Volunteering

Regional Network Coordinator (North-West England)
England, Oct 2025 - Aug 2026

Commonwealth Scholarship Commission, UK

- Coordinated four major events for the region that provided opportunities for interaction and active engagement amongst the Commonwealth Scholars in the region.
- Liased between the scholars in the region and the CSC staff through information dissemination and online group management.

Survey Participant - Research
Remote, Jan 2022 - Present

Robotics related surveys

- Online study on HRI conducted by the University of Naples Federico II in 2024.
- “Formal Robotics Education Programs: Best Practices and Future Opportunities” for the IROS 2023 forum.

Scholarships and Achievements

Master's - Commonwealth Scholarship Commission UK; Among 1% of candidates awarded a fully funded scholarship for Master's study.
Bachelor's - Crosswise Works Netherlands; Emerged top 8 out of 228 young Nigerians creating technologies to mitigate food loss.

Completed Projects

- An Adaptive Monocular Vision-Based Human Position Tracking in Dorsal Following Autonomous Wheeled Robot
- Safety-Aware Robot Navigation: Collision Avoidance in Dynamic Environment using RL
- Optimized Communication System Design for Unmanned Ground Robots
- Neural Network Algorithm-Based (LSTM, FNN, RNN) Inverse Kinematics Performance for 6-DoF Robot Arm
- Development of RobotSphere Navigation in Unstructured Environment using Unity 3D

Research Conferences Attended

- **Attendee** - IEEE International Conference on Robotics and Automation (ICRA), Philadelphia 2022.

Publications

- Okoh C.O., Buckley N. and Secco E.L. (2026): Performing the Inverse Kinematics of an ABB 6-DoF Robotic Arm with LSTM, Feedforward and Recurrent Neural Networks, Journal of Artificial Intelligence & Control Systems **(Accepted)**